```
plt.fill_between(
    xs.flatten(),
    predictive_upper,
    predictive_lower,
    alpha=0.3,
    label="95% CI"
)
```

Finally, we plot the individual samples:

```
for i in range(samples.shape[0]):
    plt.plot(xs, samples[i, :], alpha=0.5)
```

This code will produce the plot in Figure 2.17.

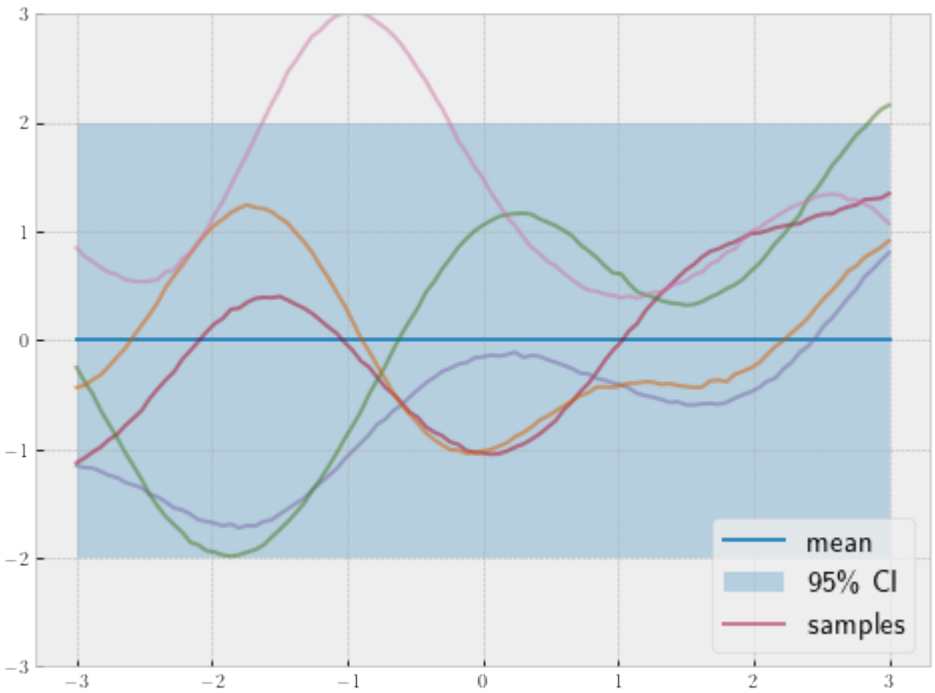


Figure 2.17 Predictions made by a prior GP with a zero mean and RBF kernel. While the mean and CI are constant, individual samples exhibit complex, non-linear behaviors.

We see that our prior GP produces a mean function that is constant at zero across the domain, and our 95% CI is also constant. This is to be expected, as we used a `gpytorch.means.ZeroMean()` object to implement the mean function, and without any training data, our prior predictions default to this zero value.

With that said, the mean and the CI are only measurements of expectation: they denote the average behavior of our predictions across many many different realizations of what could be. When we draw individual samples, however, we see that each of these samples has a very complex shape that is not at all constant. All of this is to say that while the expected value of our prediction at any point is zero, there is a wide range of values that it could take. This demonstrates that a GP can model complex, non-linear behaviors in a flexible manner.

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